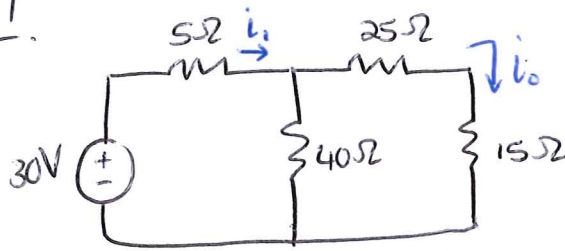


Tutorial 4 - Memo

4.1.

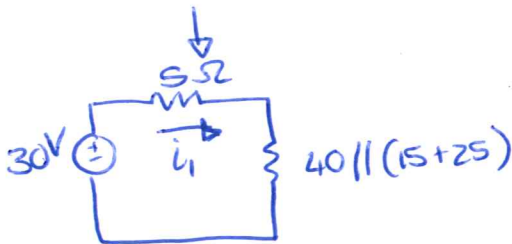


- Calculate i_0 .
- What value of input voltage will make $i_0 = 5A$?

$$* 40 \parallel (25 + 15) = 20\Omega$$

$$* i = \frac{30}{(5 + 20)} = 1.2A$$

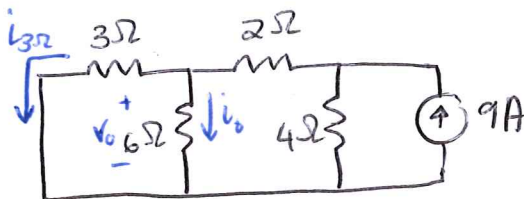
$$\therefore i_0 = i \left(\frac{40}{40 + 40} \right) = 0.6A$$



$\therefore$ Use linearity to determine V_{in} for $i_0 = 5A$.

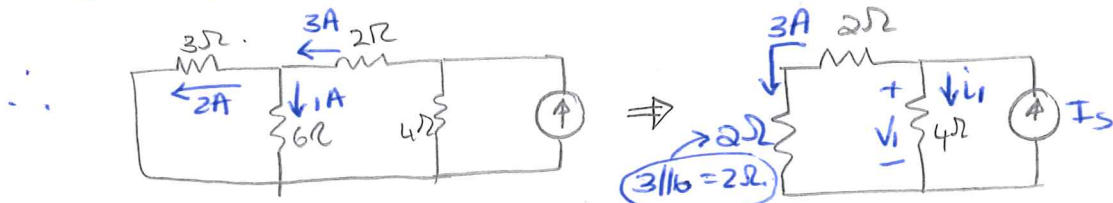
$$\Rightarrow \begin{matrix} 0.6R = 30 \\ 5R = V_{in} \end{matrix} \Rightarrow V_{in} = \frac{30 \times 5}{0.6} = 250V$$

4.4



Use linearity to determine i_0 .

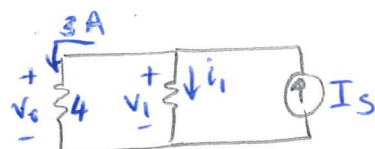
* If $i_0 = 1A$; then $v_0 = 6V$ and $i_{3\Omega} = 2A$



Hence, if:

$$I_s = 6A \rightarrow I_0 = 1A$$

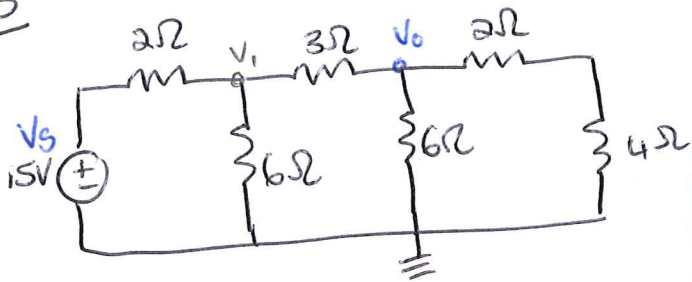
$$I_s = 9A \rightarrow I_0 = \frac{9}{6} = 1.5A$$



$$\therefore v_0 = 3(4) = 12V$$

$$\Rightarrow I_s = 3 + 3$$

4.5



Assume $V_0 = 1V$ and use linearity to find actual value for V_0 .

→ If $V_0 = 1V$:

$$\begin{aligned} \textcircled{1} \quad \frac{V_0}{6} + \frac{V_0}{6} + \frac{V_0 - V_1}{3} &= 0 \\ 2V_0 + 2V_0 - 2V_1 &= 0 \\ 2V_1 &= 4(1) \\ V_1 &= 2V \end{aligned}$$

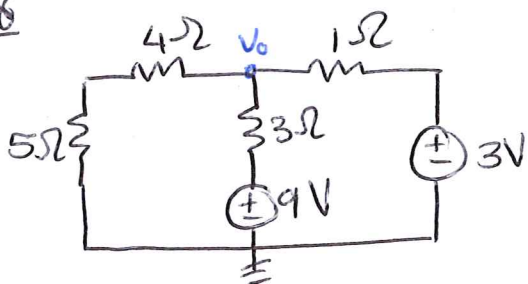
$$\begin{aligned} \textcircled{2} \quad \therefore \frac{V_1 - V_s}{2} + \frac{V_1}{6} + \frac{V_1 - V_0}{3} &= 0 \\ 3(2) - 3V_s + (2) + 2(2) - 2(1) &= 0 \\ 10 &= 3V_s \\ V_s &= \frac{10}{3} \end{aligned}$$

⇒ Thus,

$$V_s = \frac{10}{3} \rightarrow V_0 = 1$$

$$V_s = 15 \rightarrow V_0 = \frac{15}{(10/3)} = 4.5V$$

4.8



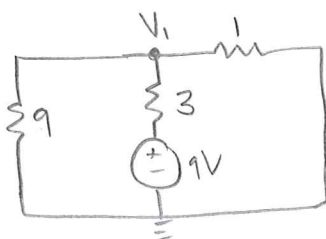
Use SUPERPOSITION to find V_0 .

* Independent sources *

(V_1 and V_2 are due to 3V- and 9V-sources respectively)

$$\Rightarrow V_0 = V_1 + V_2$$

For V_1 :

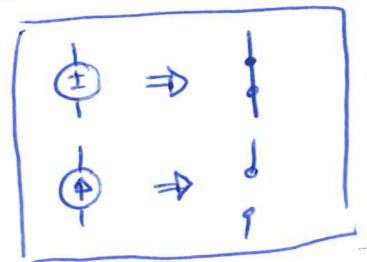


$$\frac{V_1}{9} + \frac{V_1 - 9}{3} + \frac{V_1}{1} = 0$$

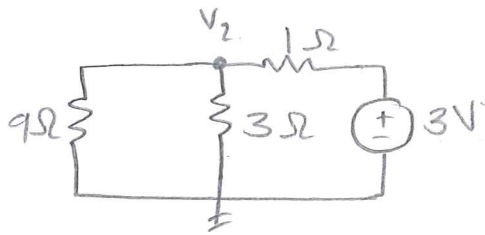
$$V_1 + 3V_1 - 27 + 9V_1 = 0$$

$$13V_1 = 27$$

$$V_1 = 2.077V$$



For V_2 :



$$\frac{V_2}{9} + \frac{V_2}{3} + \frac{V_2 - 3}{1} = 0$$

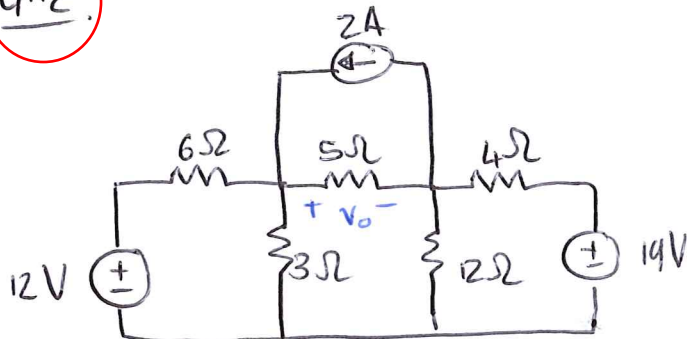
$$V_2 + 3V_2 + 9V_2 = 27$$

$$13V_2 = 27$$

$$V_2 = 2,077 \text{ V.}$$

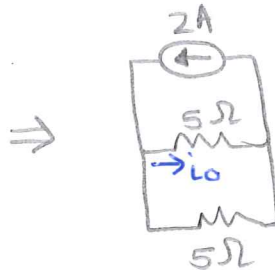
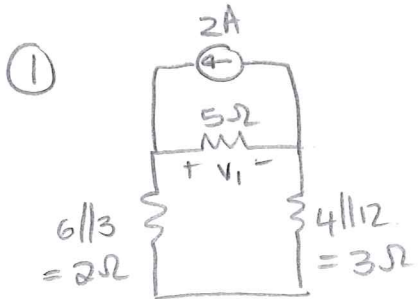
$$\begin{aligned} \therefore V_0 &= V_1 + V_2 \\ &= 2,077 + 2,077 \\ &= 4,15 \text{ V} \end{aligned}$$

4.12.



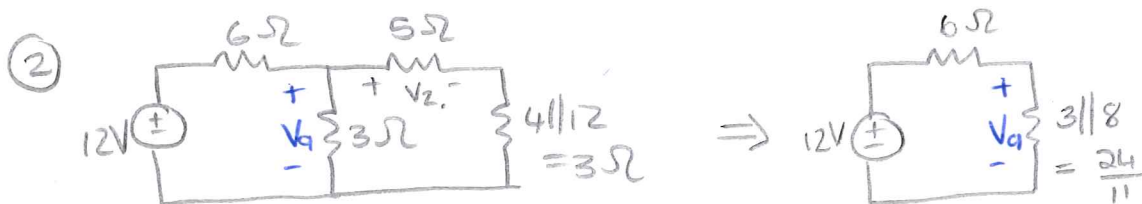
Determine V_0 using SUPERPOSITION.

$$\left[\begin{array}{l} * 3 \text{ independent sources} \\ \Rightarrow V_0 = V_1 + V_2 + V_3 \end{array} \right]$$



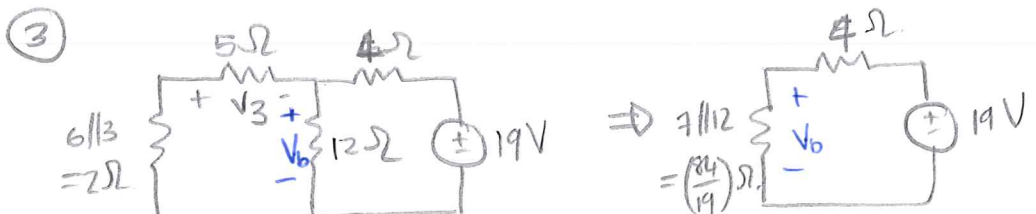
$$I_0 = 2 \left(\frac{5}{5+5} \right) = 1 \text{ A}$$

$$\therefore V_1 = 5(1) = 5 \text{ V}$$



$$* V_2 = 12 \left(\frac{24/11}{6 + 24/11} \right) = 3,2$$

$$\therefore V_2 = V_2 \left(\frac{5}{5+3} \right) = 2 \text{ V}$$

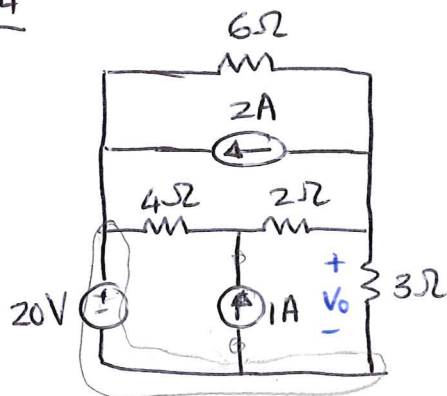


$$V_b = 19 \left(\frac{84/19}{84/19 + 4} \right) = 9.975$$

$$\therefore V_3 = V_b \left(\frac{-5}{5+2} \right) = -7.125$$

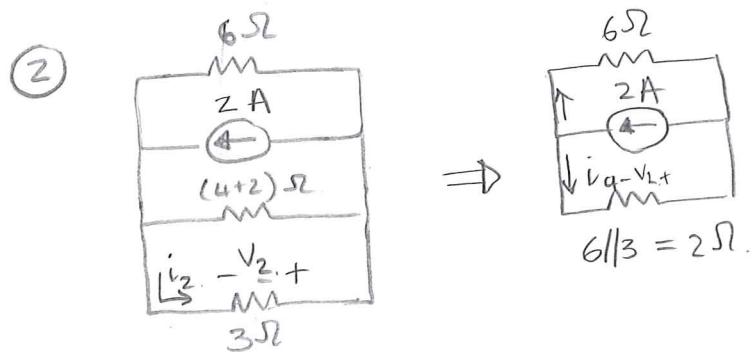
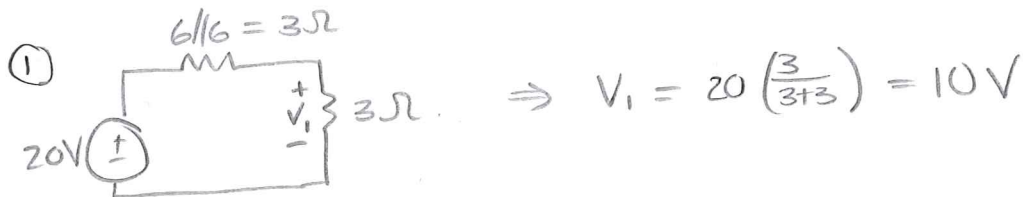
$$\therefore V_o = 5 + 2 - 7.125 = -0.125 \text{ V}$$

4.14



Use SUPERPOSITION to determine V_o

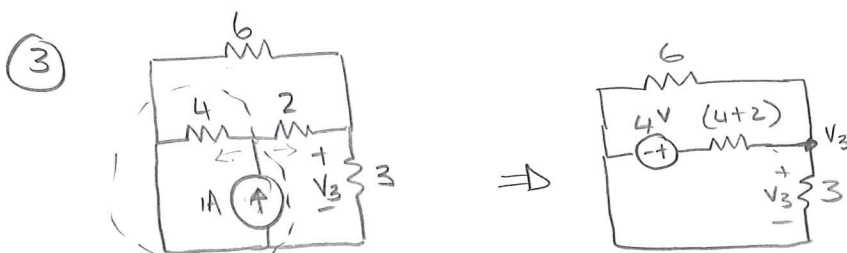
$$V_o = V_1 + V_2 + V_3$$



$$* i_q = 2 \left(\frac{6}{6+2} \right) = 1.5 \text{ A}$$

$$\therefore i_2 = 1.5 \left(\frac{6}{6+3} \right) = 1 \text{ A}$$

$$\therefore V_2 = 3(-1) = -3 \text{ V}$$



$$\frac{V_3}{3} + \frac{V_3 - 4}{6} + \frac{V_3}{6} = 0$$

$$2V_3 + V_3 - 4 + V_3 = 0$$

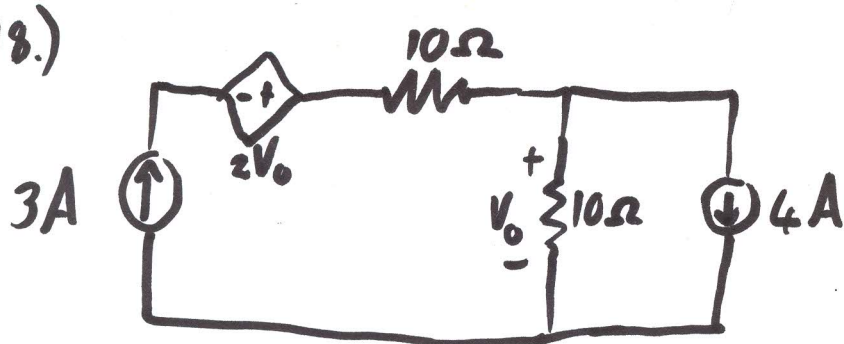
$$V_3 = 4$$

$$V_3 = 1 \text{ V}$$

SOURCE TRANSFORMATION

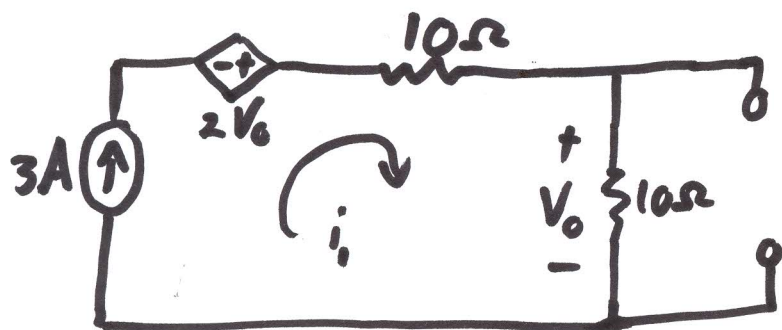
$$\Rightarrow V_o = 10 - 3 + 1 = 8 \text{ V}$$

4.18.)



Use superposition to find V_o .

① For only the 3A current source:



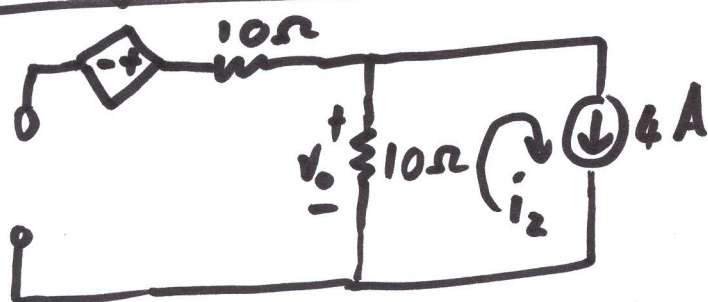
$$i_1 = 3A$$

$$\therefore V_o = i_1 (10)$$

$$V_o = (3)(10)$$

$$V_o = 30V \rightarrow$$

② For only the 4A current source:



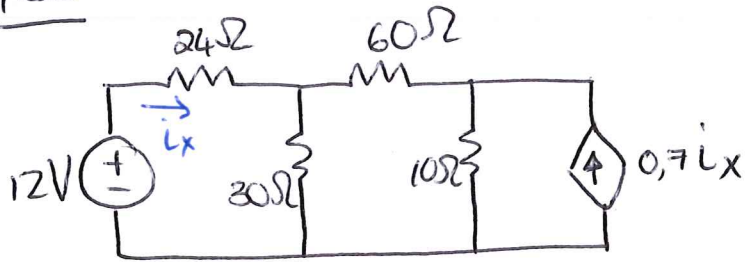
$$i_2 = 4A$$

$$-V_o = i_2 (10)$$

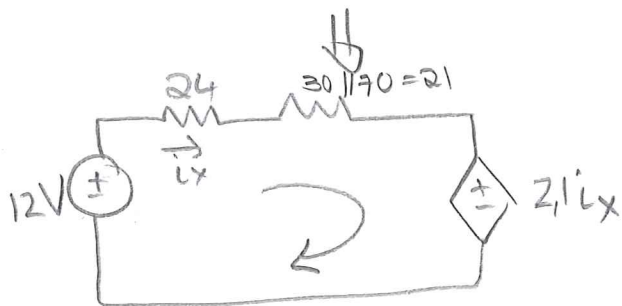
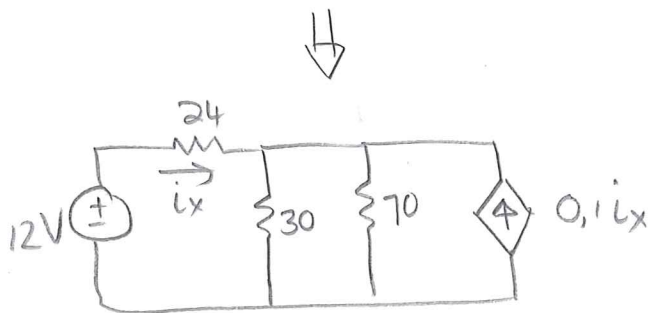
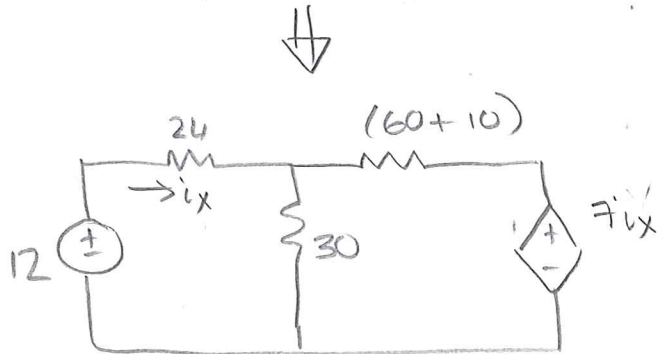
$$V_o = -40V \rightarrow$$

$$\begin{aligned} \textcircled{3} \quad V_o &= V_{o_1} + V_{o_2} \\ V_o &= 30 + (-40) \\ V_o &= -10V \end{aligned}$$

4.30



Find i_x using
Source transformation.



$$\therefore -12 + 24i_x + 21i_x + 2.1i_x = 0$$

$$47.1 i_x = 12$$

$$i_x = 0.25 \text{ A}$$



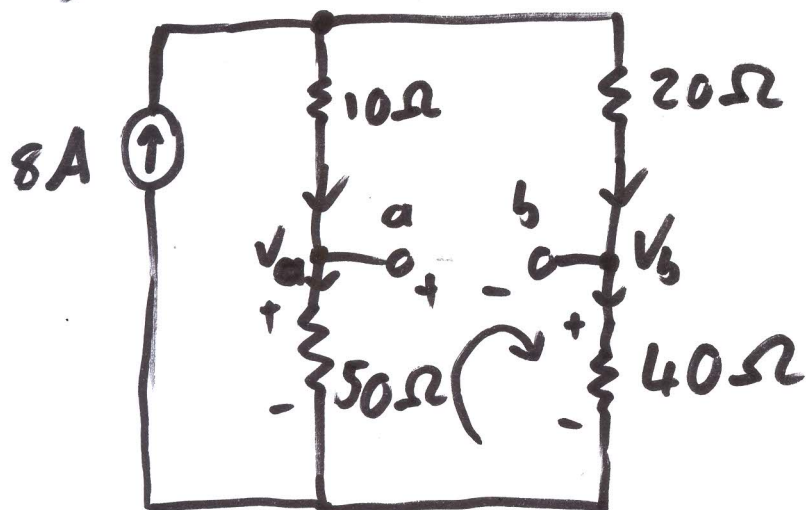
⑤ KVL $i_2: (V_{ab} = V_{th})$

$$-V_2 + (V_2 - V_3) + V_{th} = 0$$

$$-(4) + 80 + V_{th} = 0$$

$$\underline{V_{th} = -84V} \rightarrow$$

4.59.)

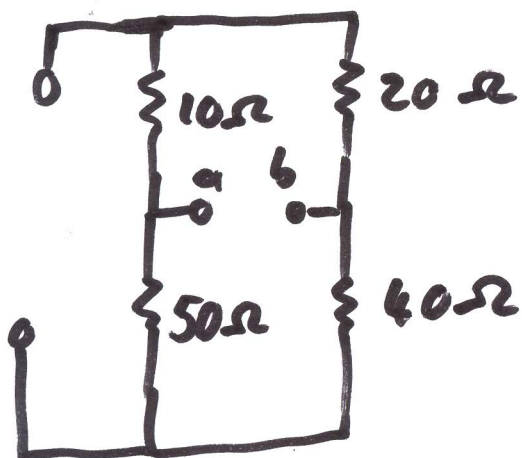


$$R_{th} = ?$$

$$V_{th} = ?$$

$$I_N = ?$$

R_{th} :



$$\begin{aligned} R_{th} &= (10+20) \parallel (50+40) \\ &= 30 \parallel 90 \\ &= 22,5 \Omega \end{aligned}$$

V_{th} :

KCL at node V_1 :

$$8 = \frac{V_1}{(10+50)} + \frac{V_1}{(20+40)}$$

$$8 = \frac{V_1}{60} + \frac{V_1}{60}$$

$$8 = \frac{V_1}{30}$$

$$\underline{V_1 = 240V}$$

Voltage division:

$$V_a = (240) \left(\frac{50}{10+50} \right)$$

$$\underline{V_a = 200V}$$

$$V_b = (240) \left(\frac{40}{20+40} \right)$$

$$\underline{V_b = 160V}$$

KVL:

$$-V_a + V_{ab} + V_b = 0$$

$$V_{ab} = V_a - V_b$$

$$V_{ab} = 200 - 160$$

$$V_{ab} = 40V$$

$$\therefore \underline{V_{th} = V_{ab} = 40V}$$

$$\begin{aligned} I_N &= \frac{V_{th}}{R_{th}} \\ &= \frac{40}{22,5} \\ &= \underline{1,78 A} \end{aligned}$$

4.20 $V = 22V$

$R = 4 \text{ Ohm}$

4.24 $V_x = 2.978V$

4.39 $R_{th} = 20 \text{ Ohm}$

$V_{th} = -84V$

4.40 $R_{th} = 2857.14 \text{ Ohm}$

$V_{th} = 60V$